

Lightweight Explainable DDoS Detection Using Optimized Feature Selection and Random Forest

Problem Statement

Distributed Denial-of-Service (DDoS) attacks overwhelm network services with malicious traffic, making them unavailable to legitimate users. Machine-learning-based approaches can detect such attacks, but many models rely on a large number of network traffic features. This can increase computational requirements and make the model's decisions difficult to interpret.

Proposed Idea

We propose a **lightweight and explainable machine-learning framework for DDoS detection** using **optimized feature selection, Random Forest, and SHAP (SHapley Additive exPlanations)**.

The main objective is to investigate whether a significantly smaller subset of network traffic features can detect DDoS attacks with performance comparable to a model using the complete feature set.

The proposed pipeline is:

DDoS Dataset → Data Preprocessing → Feature Selection → Random Forest → DDoS/Normal Classification → SHAP Explainability

Multiple feature-selection techniques can be evaluated to identify the most informative network features. A Random Forest model trained using all available features will first serve as the baseline. The model will then be retrained using progressively smaller optimized feature subsets.

The models will be compared using metrics such as **precision, recall, F1-score, false-positive rate, inference time, model size, and number of features**. This will allow us to study the trade-off between detection performance and computational efficiency and determine a suitable minimal feature set.

Explainability

SHAP will be incorporated to explain the Random Forest predictions. It can provide:

- **Global explanations:** Which network features contribute most to DDoS detection overall.
- **Local explanations:** Why a particular network flow was classified as normal or malicious.

SHAP analysis can also help investigate whether the features retained by the feature-selection process are consistently important to the model.

Research Objective

The central research question is:

Can we significantly reduce the number of network traffic features required for Random Forest-based DDoS detection while maintaining strong detection performance, low computational overhead, and interpretable predictions?

Rather than proposing another ML classifier for DDoS detection, the research focuses on finding an effective **performance–efficiency–explainability trade-off**.

Expected Contribution

The expected outcome is a lightweight DDoS detection framework that:

1. Uses substantially fewer network features.
2. Maintains competitive DDoS detection performance.
3. Reduces computational and inference requirements.
4. Provides interpretable predictions using SHAP.
5. Identifies the most relevant and stable network characteristics associated with DDoS traffic.

The work can initially be evaluated using established public DDoS datasets, making the project feasible without deploying or generating attacks on real-world networks.

The primary contribution would therefore be **optimized feature selection and explainability**, while Random Forest acts as the core detection model.